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SHANGHAI JIAO TONG UNIVERSITY

研究生学位论文开题报告

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研究课题来源 Source of Research Project	请在合适选项前画✓ Please select proper options by “✓” . ☐ 国家自然科学基金课题 NSFC Research Grants ☐ 国家社会科学基金 National Social Science Fund of China ☐ 国家重大科研专项 National Key Research Projects ☐ 其它纵向科研课题 Other Governmental Research Grants ☐ 企业横向课题 R&D Projects from Industry ☒ 自拟课题 Self-proposed Project ☐ 其它 Other [_____]

1、请综述课题国内外研究进展、现状、挑战与意义，可分节描述。博士生不少于

10,000 汉字，硕士生不少于 5,000 汉字。请在文中标注参考文献。 Please review the frontier, current status, challenges and significance of the research topic. The citations should be marked in the context and listed in order at the end of this section. No less than 8,000 words for doctoral students and 4,000 words for master students if written in English.

[1.1 Background and motivation

Large language model (LLM) agents - systems in which an LLM plans, calls tools, and acts over several steps - have moved quickly from research demonstrations to enterprise deployment [1]. A natural extension is the multi-agent system (MAS), in which several specialised agents (for example a planner, a coder, and a reviewer) collaborate on one task. Popular frameworks such as AutoGen [2], MetaGPT [3], CAMEL [4] and ChatDev [5] show that role specialisation and structured communication can, in principle, decompose complex problems. In business settings this promise is concrete: firms increasingly build self-service analytics agents so that non-technical staff in sales, operations and finance can answer their own data questions instead of queuing for a human analyst. The present proposal is motivated directly by this trend, which the author observes first-hand while building such systems in a corporate-and-investment-banking analytics team.

1.2 The central puzzle: more agents do not reliably mean better results

Despite the enthusiasm, the empirical record is mixed: multi-agent systems frequently fail to outperform a single, well-prompted agent, and sometimes do worse [6,7]. A recent large-scale study, MAST (Why Do Multi-Agent LLM Systems Fail?), annotated more than 1,600 execution traces across seven frameworks and produced the first Multi-Agent System Failure Taxonomy: fourteen failure modes grouped into three categories - (i) specification and system-design issues, (ii) inter-agent misalignment, and (iii) task verification and termination [7]. Importantly, failures are spread across all three categories and call for design-level fixes rather than larger models. Work on multi-agent debate reaches a similar caution: reported gains may stem largely from aggregating multiple answers rather than from the interaction itself [8,9]. Together these results reframe the question from do more agents help? to when, and why, do they fail?

1.3 Verification as a likely bottleneck

One failure category is especially relevant when tasks have checkable answers. A growing literature shows that LLMs are weak at verifying and correcting their own reasoning: they often cannot reliably locate their own errors [10], and iterative self-feedback methods such as Reflexion [11] and Self-Refine [12] help only under specific conditions, with smaller models in particular needing an external verifier. Evaluation itself increasingly relies on the LLM-as-a-judge paradigm, which matches human preferences at over 80% agreement but carries known biases [13]. In a multi-agent pipeline, weak verification means a plausible-but-wrong intermediate result can propagate unchecked into the final answer - the mechanism the MAST taxonomy labels no/incomplete verification and incorrect verification [7].

1.4 Business analytics as an ideal but under-used testbed

Most MAS evaluations use coding, mathematics or open-ended research tasks [6,7]. Business analytics is an attractive yet largely unused testbed because its tasks have objective ground truth: a question such as which region had the highest quarter-over-quarter revenue growth? resolves to a specific number or SQL result that can be checked automatically. A mature set of public benchmarks already supplies such tasks - Spider [14] and the database-grounded BIRD benchmark [15] for text-to-SQL, the enterprise-scale Spider 2.0 [16], the data-analysis agent benchmark InfiAgent-DABench [17], TableBench [18] for table question answering, and the financial reasoning datasets FinQA [19] and TAT-QA [20]. However, these benchmarks evaluate single agents and score only final-answer accuracy; none diagnoses multi-agent failure modes, and none isolates the role of verification.

1.5 Research gap and significance

This leaves a clear gap at the intersection of three literatures: multi-agent systems and their failures [6,7], verification and self-correction [10,11,12], and verifiable analytics benchmarks [15,17]. No existing work (a) measures whether multi-agent systems beat single agents specifically on business-analytics tasks, (b) profiles where their failures fall using the MAST taxonomy, or (c) tests whether adding an explicit verification mechanism closes the gap. Filling this gap is scientifically significant - it would yield the first failure-mode profile of MAS in a verifiable business domain and a reusable, verification-aware benchmark - and practically significant, giving firms that deploy self-service analytics agents an evidence-based way to measure quality and to decide when multi-agent designs are worth their additional cost.]

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2、课题研究目标、主要研究内容和拟解决的关键问题。 Research objectives, main contents and key issues to be solved.

[Research objective

The objective is to build a verification-aware benchmark for multi-agent business analytics and to use it to (a) measure when multi-agent systems outperform single agents, (b) quantify how their failures distribute across the MAST categories, and (c) test whether an explicit verifier reduces failures and closes the single-versus-multi gap.

Research questions and hypotheses

RQ1 - Do multi-agent systems outperform single agents on business-analytics tasks, and does this depend on task complexity? H1: multi-agent and single-agent performance are comparable on simple queries, but multi-agent systems gain an advantage on complex, multi-step tasks.

RQ2 - How do multi-agent failures distribute across the MAST categories in analytics? H2: a disproportionate share of failures fall in the Task-Verification category, relative to the coding/math mix reported by MAST, because numeric answers can be plausible yet wrong.

RQ3 - Does an explicit verification mechanism improve outcomes? H3: adding a dedicated verifier agent and an executable check (re-running the SQL or recomputing the metric) reduces verification failures and closes most of the single-versus-multi gap.

Main contents

- (1) A curated business-analytics task suite with ground-truth answers and intermediate

milestones. (2) An experimental comparison of single- and multi-agent configurations across task-complexity levels. (3) An automated failure-annotation pipeline applying the MAST taxonomy. (4) A controlled verifier intervention.

Key issues to be solved

(i) Defining business-analytics tasks that are both realistic and automatically checkable, with milestone checkpoints for partial credit. (ii) Obtaining reliable, scalable failure annotation by validating an LLM-as-judge annotator against a hand-labelled subset. (iii) Cleanly isolating the verification effect while controlling for confounds such as model choice, token cost, and task difficulty.]

3、拟采取的研究方法、研究方案及其可行性分析。 Research methods and research scheme to be adopted and feasibility analysis.

[Data and task construction

The benchmark will be assembled from existing public datasets with verifiable answers - BIRD [15], Spider 2.0 [16], InfiAgent-DABench [17], TableBench [18], and FinQA/TAT-QA [19,20] - re-cast into a unified business-analytics suite covering sales, finance and operations questions. Each task carries a gold answer for automatic execution-based scoring and is decomposed into milestones (for example correct table/column selection, correct aggregation, correct final value) to enable milestone-based KPIs in the style of MultiAgentBench [6]. No proprietary or personal data is used.

System implementation

Single-agent and multi-agent configurations will be implemented on an open framework, building on MARBLE and AutoGen [2,6]. The multi-agent conditions vary coordination topology (chain, star, graph) and role decomposition (for example planner, SQL/analysis agent, critic). The verifier condition adds a dedicated verification agent plus an executable check that re-runs generated SQL or recomputes the reported metric against the data.

Metrics and analysis

The primary outcome is task success by execution match against the gold answer; secondary outcomes are milestone-achievement rate and cost (tokens, latency). Every failed trace is labelled with the MAST taxonomy using an LLM-as-judge annotator [7,13], whose reliability is validated against a hand-labelled sample of about 150 traces with inter-rater

agreement (Cohen's kappa) reported. A factorial design crosses task complexity (low/high) x architecture (single, multi-chain, multi-graph) x verifier (off/on), with at least one small model (such as gpt-4o-mini) and one stronger model for robustness. Success is analysed with mixed-effects logistic regression (random effects for task and model); the failure-category distribution is compared with chi-square tests; the verifier effect is estimated as the change in success and in the verification-failure rate.

Feasibility

The study relies only on public data and open-source frameworks and reuses MAST's released taxonomy and annotator [7], which lowers technical risk. Compute is modest: tasks run through model APIs within a limited budget, and small models are viable subjects [6]. The author's professional experience building analytics agents reduces the engineering learning curve. The main risks - annotator noise and dataset heterogeneity - are mitigated by human validation of the annotator and by a unified task schema; scope can be narrowed (fewer datasets or one model family) if needed.]

4、课题的创新点 Novelties of the proposed topic.

[1. Domain-first. The first multi-agent benchmark and failure analysis situated in business analytics - a domain whose verifiable ground truth makes failure objectively measurable, unlike the coding, mathematics and open-ended tasks used in prior MAS evaluations [6,7].

2. Taxonomy application. The first empirical application of the MAST failure taxonomy [7] to analytics, testing the novel hypothesis that verification failures dominate when answers are checkable.

3. Intervention, not just description. Rather than only cataloguing failures, the study runs a controlled verifier intervention and quantifies how much of the single-versus-multi gap it closes.

4. Cost-aware and practitioner-facing. By reporting accuracy alongside token and latency cost, the work yields an actionable decision rule for when multi-agent designs are worth their overhead - directly useful to firms deploying self-service analytics agents.]

[Indicative schedule (about eight months)

Months 1-2: finalise the literature review; design the benchmark schema and milestone scheme. Months 3-4: build the task suite and the single-/multi-agent harness. Month 5: run

single- versus multi-agent experiments and collect traces. Month 6: failure annotation (MAST) and annotator validation. Month 7: verifier-intervention experiments and analysis. Month 8: write up and release the benchmark and code.

Expected outcomes

(1) An open-source verification-aware business-analytics benchmark (tasks, gold answers, milestones, and the failure-annotation pipeline). (2) Empirical findings on whether and when multi-agent systems beat single agents in analytics, and on how their failures distribute. (3) Evidence on the effect of an explicit verifier. (4) A completed master's thesis and, potentially, a short workshop paper and a concise practitioner guideline.]

5、与本课题有关的工作积累、已有的研究工作成绩。 Prior experience and accomplished achievements related to the proposed topic.

[The author works as a data analyst in a corporate-and-investment-banking division, where current responsibilities include building agentic and plugin-based analytics systems that let business and sales staff perform their own analyses. This provides direct, daily exposure to the problem the thesis studies - including the practical difficulty of benchmarking and comparing different agent configurations - together with working proficiency in Python and machine learning. This background supports both the engineering implementation and the practical relevance of the proposed benchmark. The proposal also builds on prior coursework in quantitative methods and data analytics within the MIB programme.]

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